

WASP-43b

R-0323

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260227 · created 2026-07-18T02:36:21+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461098.73018 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1851 +/- 0.0097
Transit depth (Rp/Rs)^2	3.42 +/- 0.36 [%]
Orbital Inclination (inc)	83.69 +/- 1.15
Airmass coefficient 1 (a1)	0.951 +/- 0.013
Airmass coefficient 2 (a2)	0.031 +/- 0.0057
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#2 - [272, 366]
Optimal Aperture	3.34
Optimal Annulus	7.35
Transit Duration (day)	0.0565 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1851 $\pm$ 0.0097 vs 0.1594 (deviation 2.65 σ)
Duration fit vs published	0.0565 d vs 0.0512 d (deviation 1.94 σ)
Mid-time O-C	6.6 min
Depth SNR	19.1
Residual scatter	1.36 %

Light curve

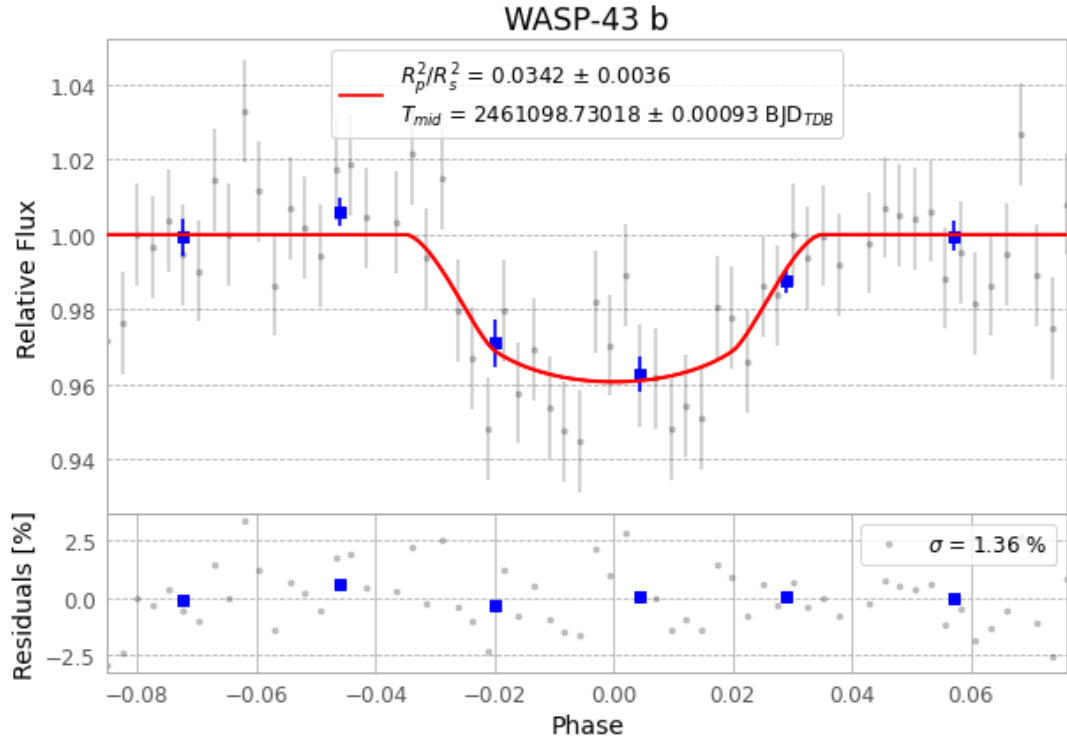


Figure 1 — FinalLightCurve_WASP-43 b_2026-02-26.png (SHA-256 fcb6d6f39829e9b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [439, 188], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ad883fa14df6da60...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

64 FITS frames (42 MB), WASP-43260227034216.FITS → WASP-43260227065117.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260227.log (a7df062d1db1...), FinalLightCurve_WASP-43 b_2026-02-26.png (fcb6d6f3982...), FinalLightCurve_WASP-43 b_2026-02-26.csv (9f715025f4d6...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 02:36Z.